

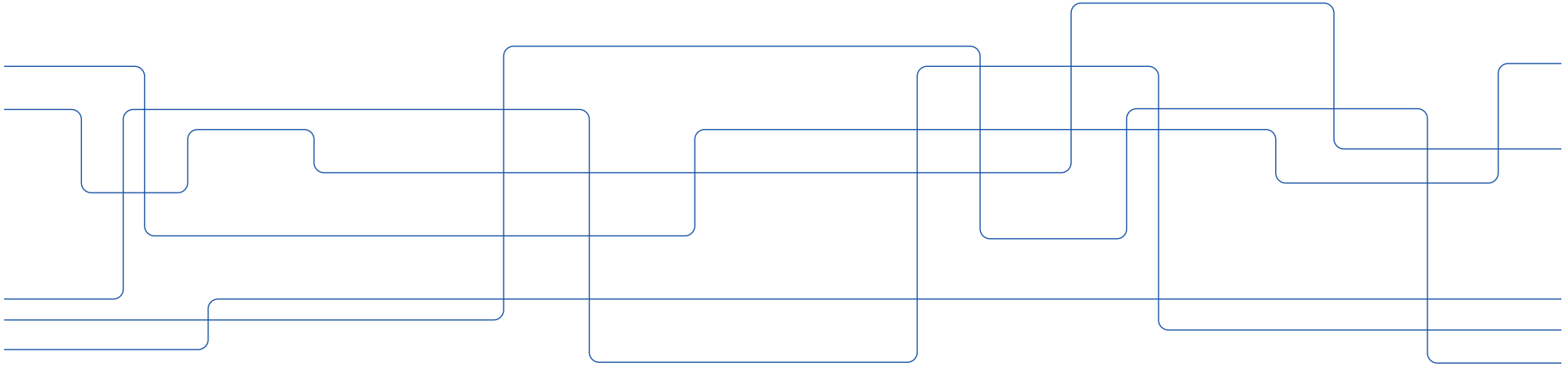
Molecular Simulations

VT25 Simulation Methods in Medical Engineering

Alessandra Villa

BioExcel Center of Excellence for Computational Biomolecular Research @ PDC Center for High Performance Computing – KTH

avilla@kth.se



Intended learning outcomes

- Recognise application cases for molecular simulation methods in the field of Medical Engineering
- Distinguish the methods for molecular simulations
- Understand the molecular models

Teaching activities and final report

- Lectures
- Computer laboratory in groups of TWO persons (self sign-up)

Groups 1 to 10 -> Thursday 30th at 13-15

Group 11 to 21 -> Thursday 30th at 15-17

- Final report (pass/not pass) – to be submitted on Canvas in PDF format 14th February 2025

Final report

- Cover all the points in the instruction.
- Concept/ideas correctly express.
- Written in a clear way. Plots/Figure with caption, legend and labels on the axis. Values together with unit.

- Deadline 14th February 2025

Literature

Books:

- Leach A. R. Molecular modelling: Principles and applications
- Schlick T. Molecular Modelling and Simulation: An Interdisciplinary Guide
- Allen & Tildesley Computer Simulation of Liquids
- H.J.C. Berendsen “Simulating the physical world”

Articles:

- Karplus and McCammon “Molecular dynamics simulation of biomolecules” Nat. Struct. Biol. 9 (2002) p 646
- H.J.C Berendsen “Bio-Molecular Dynamics comes of Age” Science 271 (1996) p 954; H.J.C Berendsen “Reality Simulation – Observe while it happens” 294 (2001) p 2304
- W.F. van Gunsteren et al. “Biomolecular Modeling: Goals, Problems, Perspectives” Angew Chem Int Ed 45 (2006) p 4064
- A. Hospital et al. “Molecular dynamics simulations: advance and applications”. Adv.Appl. Bioinform Chem. (2015)
- Perilla et al. “Molecular dynamics simulations of large macromolecular complexes.” Curr. Op. In Structural Biology (2015)

Videos:

Nobel lectures in chemistry 2013: Michael Levitt, Martin Karplus and Arieh Warshel

Day 1

Introduction & examples

Experiment vs Simulation

Molecular simulations methods

Algorithms for molecular dynamics

What are molecular simulations?

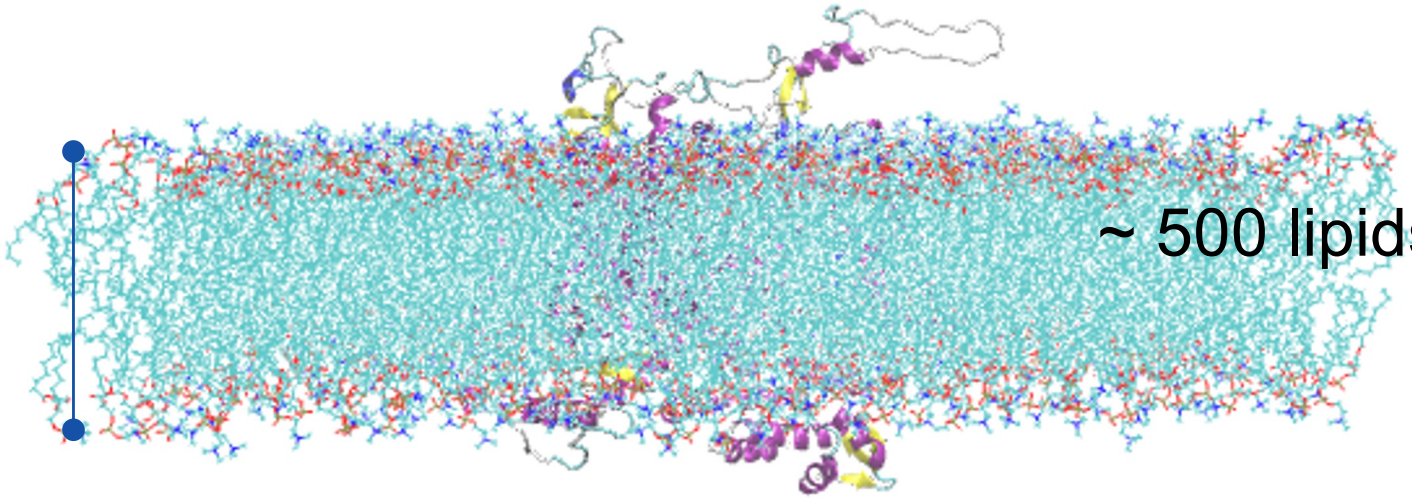
tools to mimic the behaviour of molecules and atoms, usually associated with computer modelling

Why do we simulate molecules

- Cannot see atoms

Everything you think you know about molecules is a model.

Nav1.1 (protein)
~18000 atoms



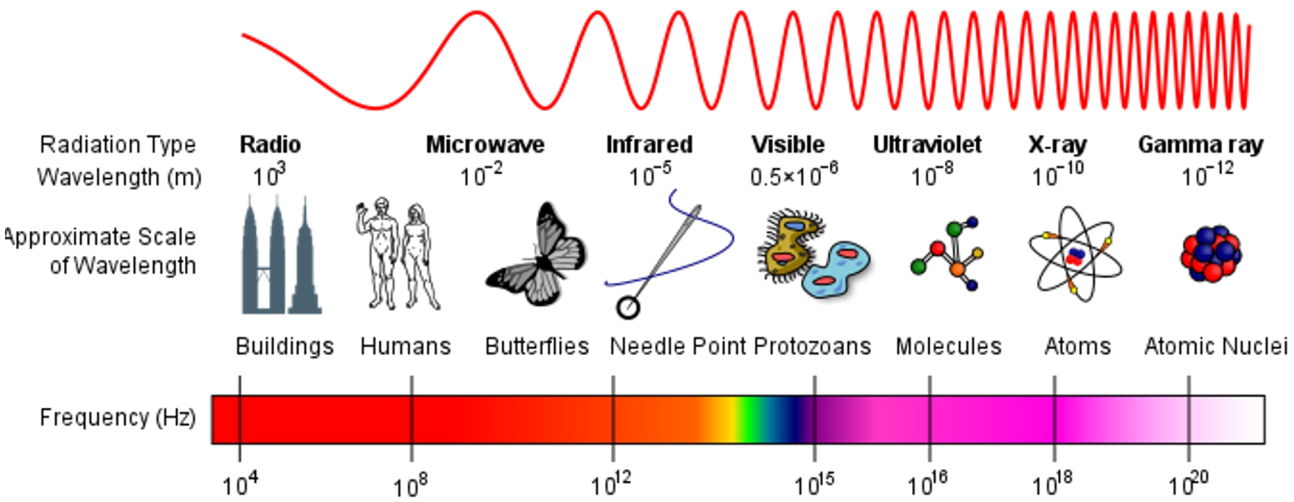
9 nm

~ 500 lipids

1 phospholipid
~ 130 atoms

nanometer (nm) = 10^{-9} m

A structure can be visualized only if light with a wavelength comparable to **its dimension** is used. (Rayleigh distance)



Licensed under CC BY-SA 3.0 via Wikimedia Commons - http://commons.wikimedia.org/wiki/File:EM_Spectrum_Properties_edit.svg#/media/File:EM_Spectrum_Properties_edit.svg

$1\ \text{Å} = 10^{-10}\ \text{m}$

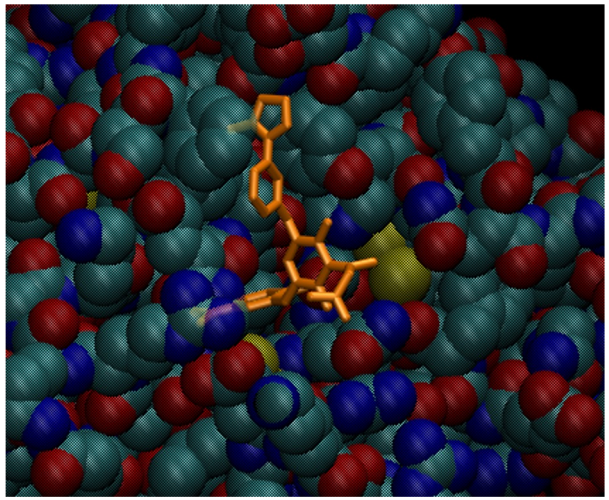
“seeing is believing”

“Unfortunately, we cannot describe at the chemical level how a molecule functions unless we first know

its structure” *James Watson (Nobel price) in a review in 1964)*

and

its dynamics



Why do we simulate biomolecules

- Not everything can be measured by biophysical tools

Some molecules do not crystallize (X-ray). Some are not soluble (NMR). Some molecules do not have enough contrast (Cryo-EM)

Techniques to get fine 3D molecular structure

- Xray crystallography

  a molecule may not crystallize

- Solution Nuclear magnetic resonance

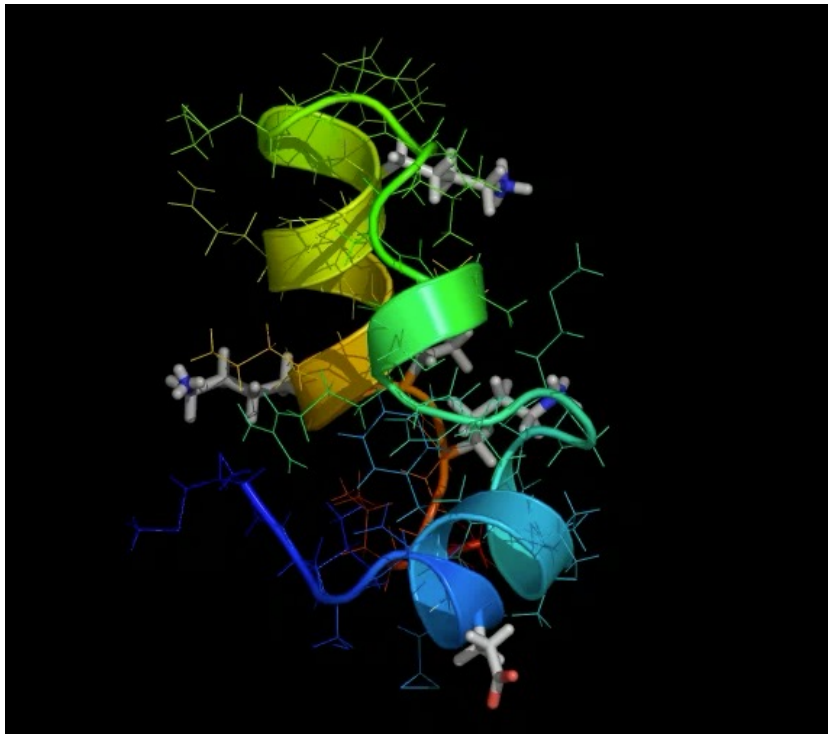
  a molecule may not be soluble in water

- Cryo-electron microscopy

  a molecule may have to low contrast to be detected

Why do we simulate biomolecules

- Cannot measure energetics or dynamics at an atomic level.



ATOMS MOVE

Molecular Simulations: what are they good for

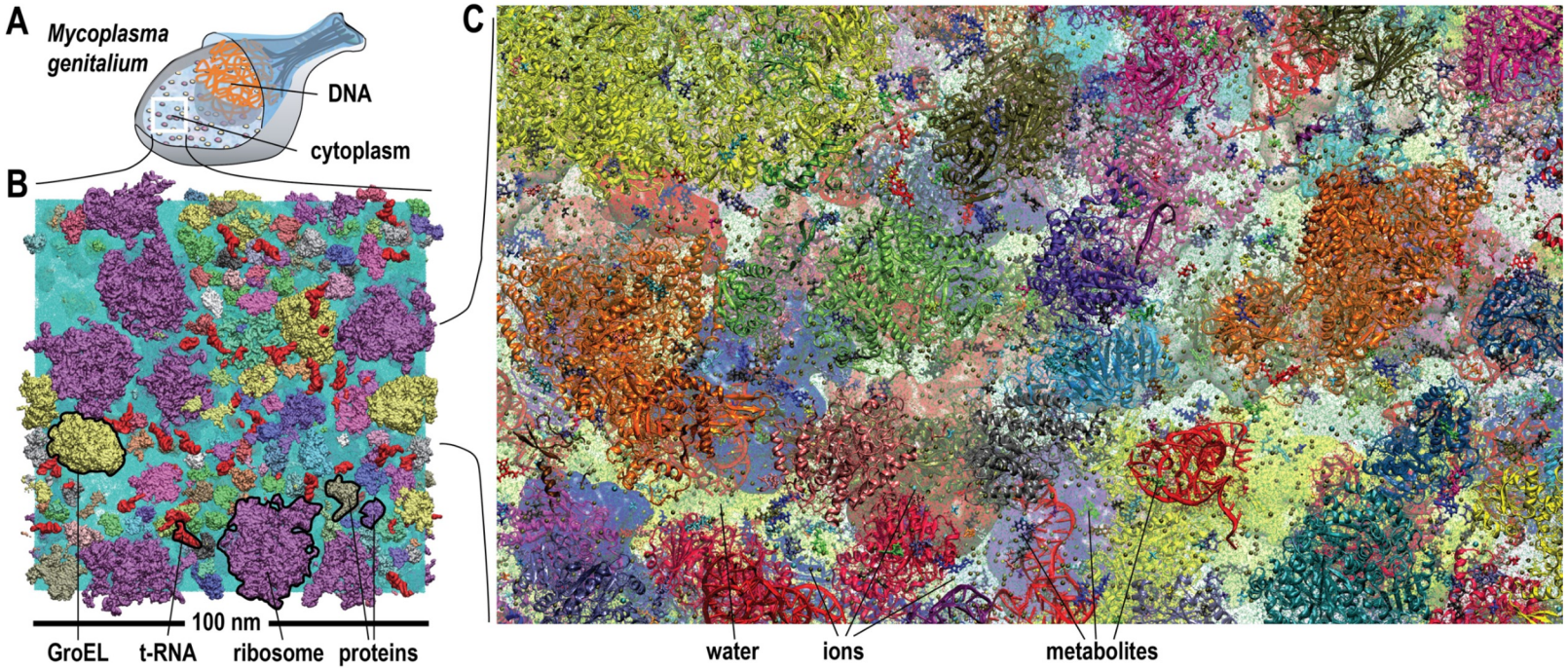
- Visualize three-dimensional structure at the atomistic level.

To show connections which are not obvious in 1D (sequence) space

- Understand what makes molecules functions (e.i. molecular recognition, molecular stability, molecular flexibility, conformational changes)
- Evaluate physical properties (e.i. diffusion, transport)
- Test hypothesis in silico (Rational drug design - Engineering of new molecular devices)

Many molecules in the cell

Molecular model of a bacterial cytoplasm.



(A) Schematic illustration of Mycoplasma genitalium (MG). (B) Equilibrated MGh system highlighted with proteins, tRNA, GroEL, and ribosomes. (C) MGh close-up showing atomistic level of detail. See also supplementary Figures 1 and 2 for structures of individual macromolecules and metabolites as well as supplementary Figure 3 for initial configurations of the simulated systems.https://doi.org/10.7554/eLife.19274.003

Molecular Simulations: what are they good for

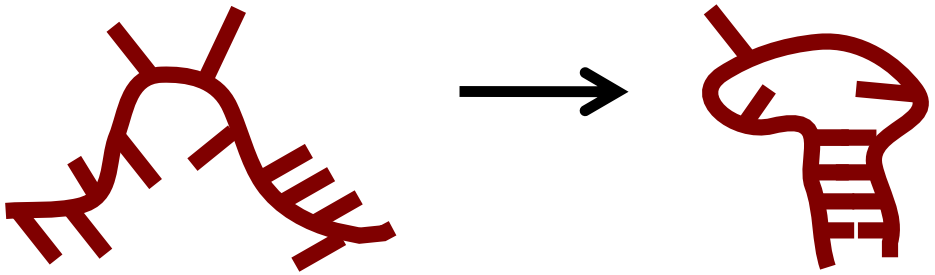
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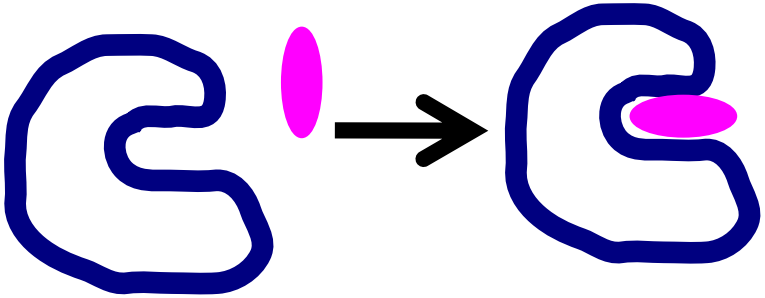
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Many processes in the cell

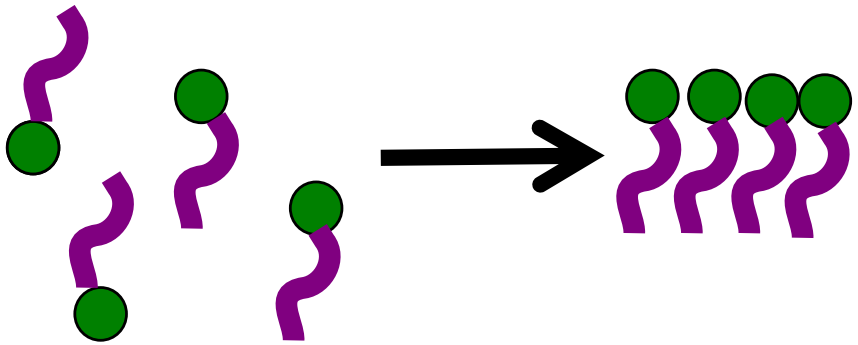
Folding



Binding



Aggregation



driven by a tiny balance of several different force

Molecular Simulations: what are they good for

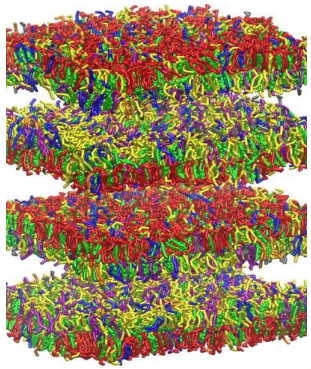
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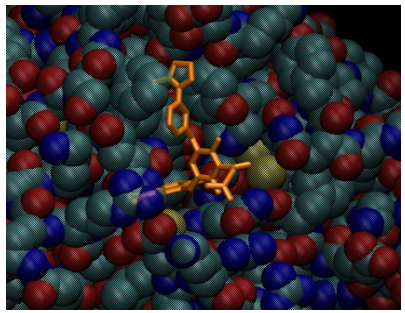
- Understand what makes molecules functions (e.i. molecular recognition, molecular stability, molecular flexibility, conformational changes)
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Questions

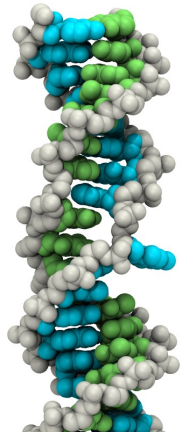
How behave complex membrane?



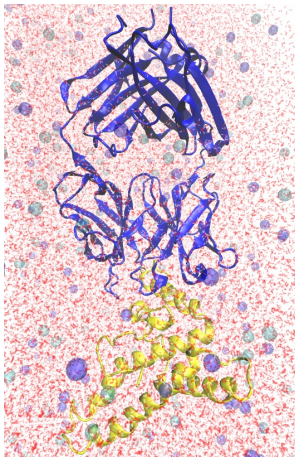
How can we inhibit Factor Xa (enzyme in the anticoagulation cascade)?



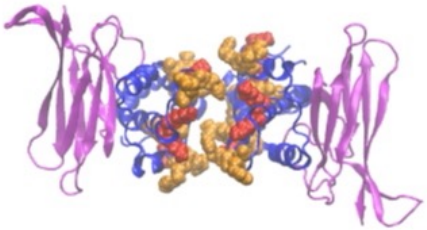
How do DNA breath?



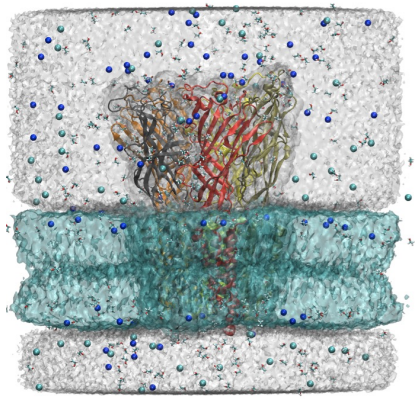
How do an antigen interact with antibody?



How do egg coat proteins work together?



How do ions channel work?



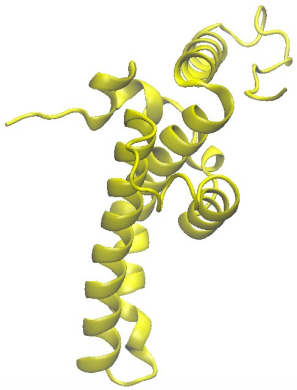
Antibody design

Antibody



as therapeutics (rheumatoid arthritis, multiple sclerosis and some types of cancer)
as detector in disease or condition diagnostics

Antigen



a unique site at the outside of the pathogen (e.i virus, bacteria)
Its presence in the body normally triggers an immune response.

Antibody

- High specificity

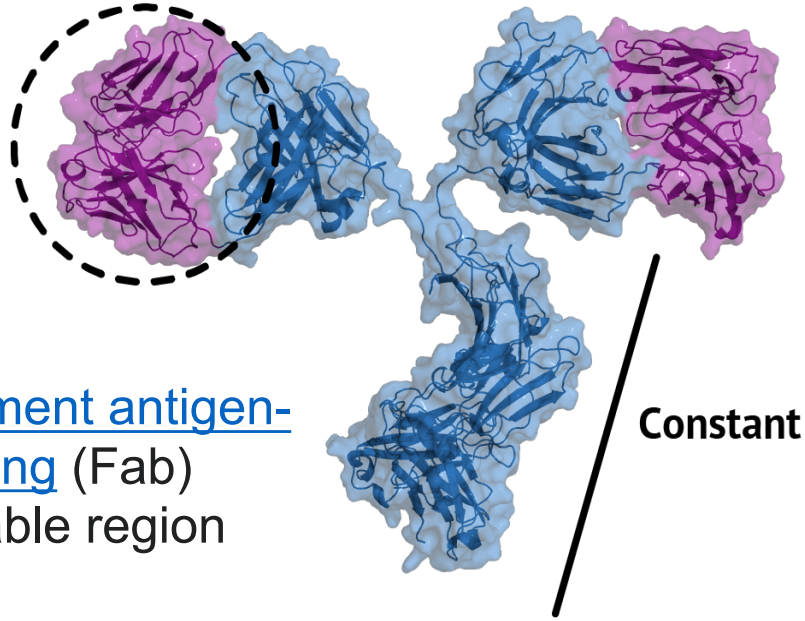
Hypervariable loops



90°

fragment antigen-binding (Fab)

Variable region (Fv)

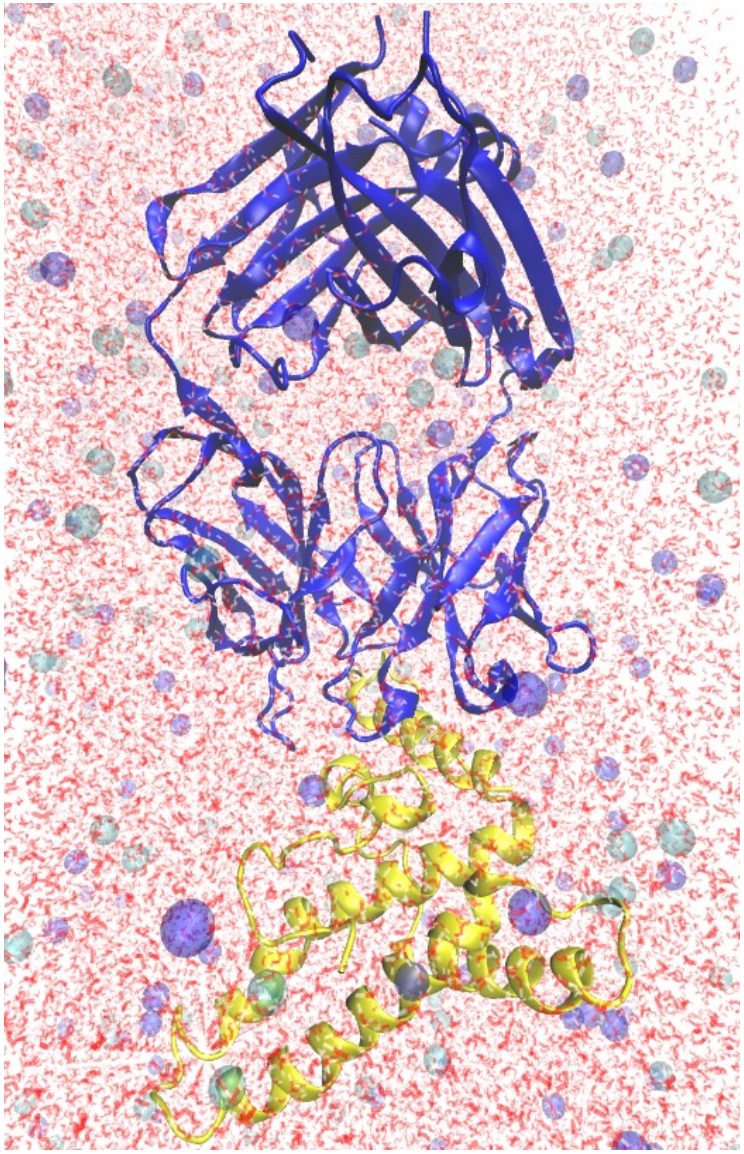


fragment antigen-binding (Fab) variable region

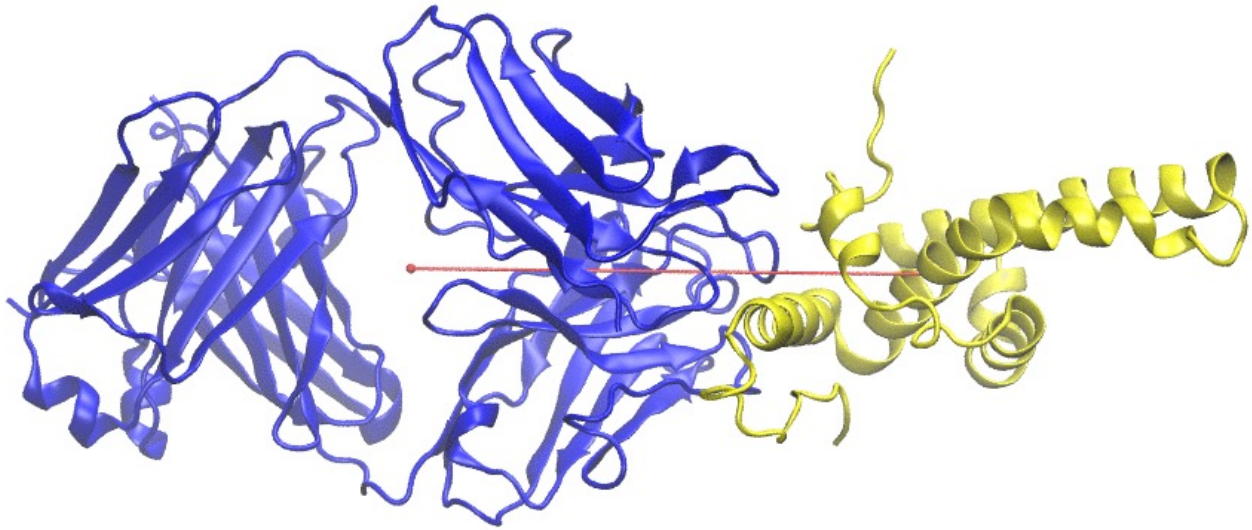
Constant region

- Low immunogenicity

How an antigen interacts with antibody



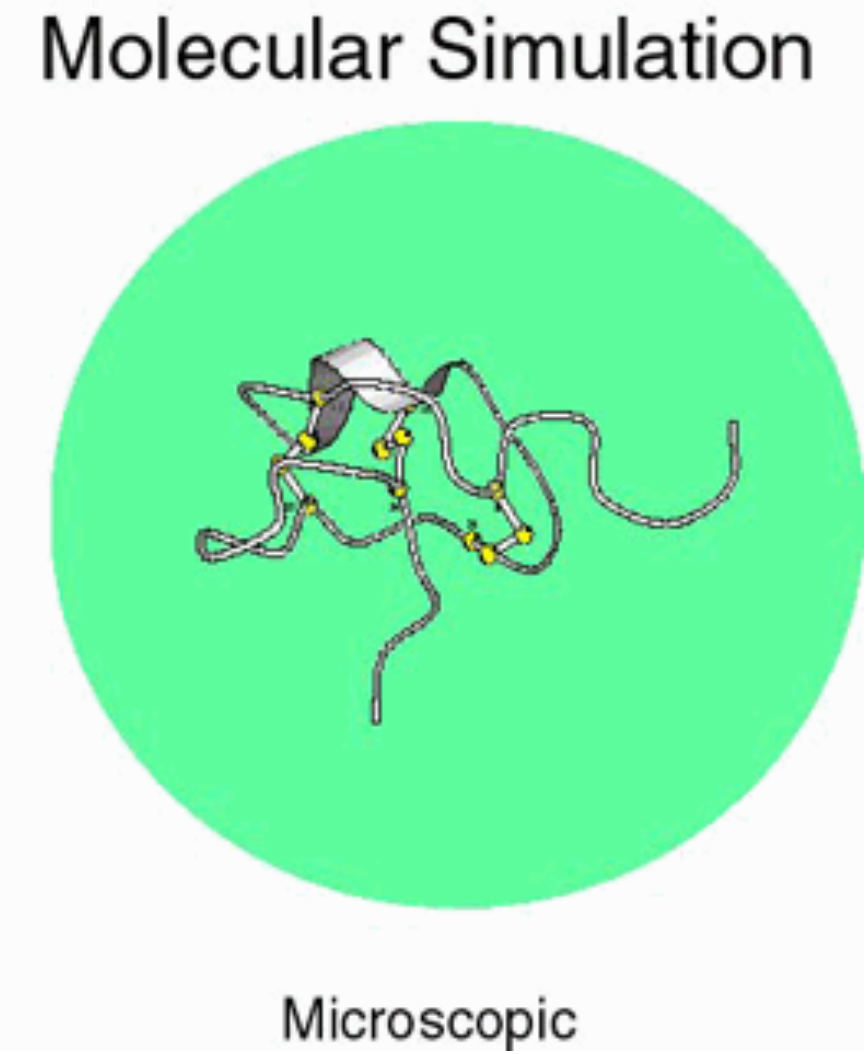
How an antigen interacts with antibody

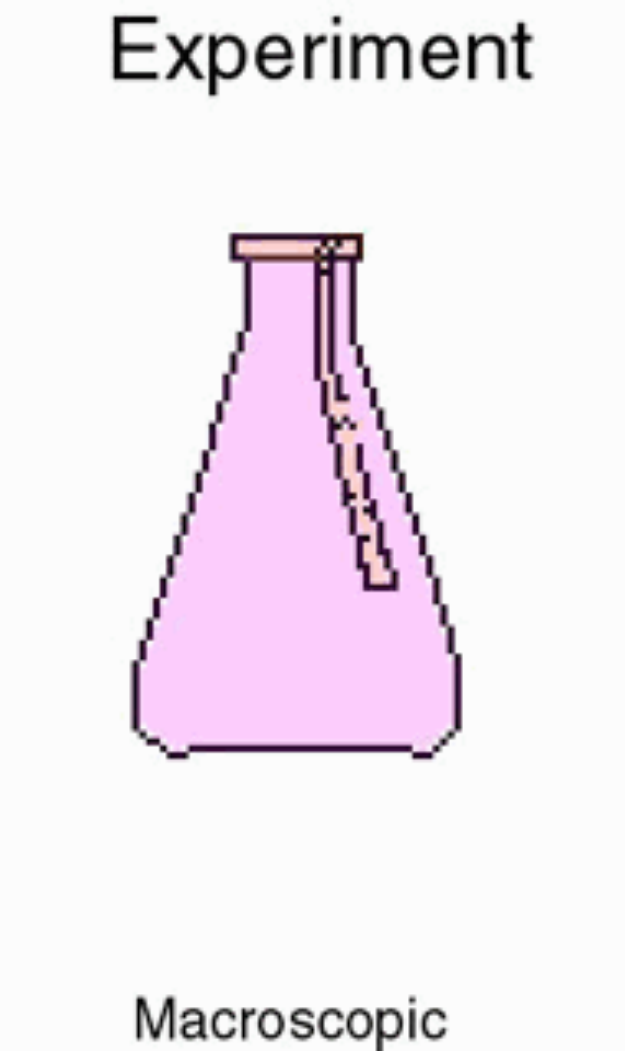


One goal of a molecular simulation

to generate enough representative conformations of the molecular system in such a way that accurate values of a property can be obtained.

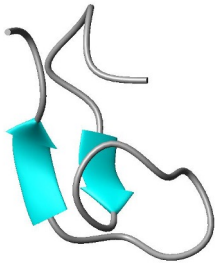
Molecular Simulation vs Experiment



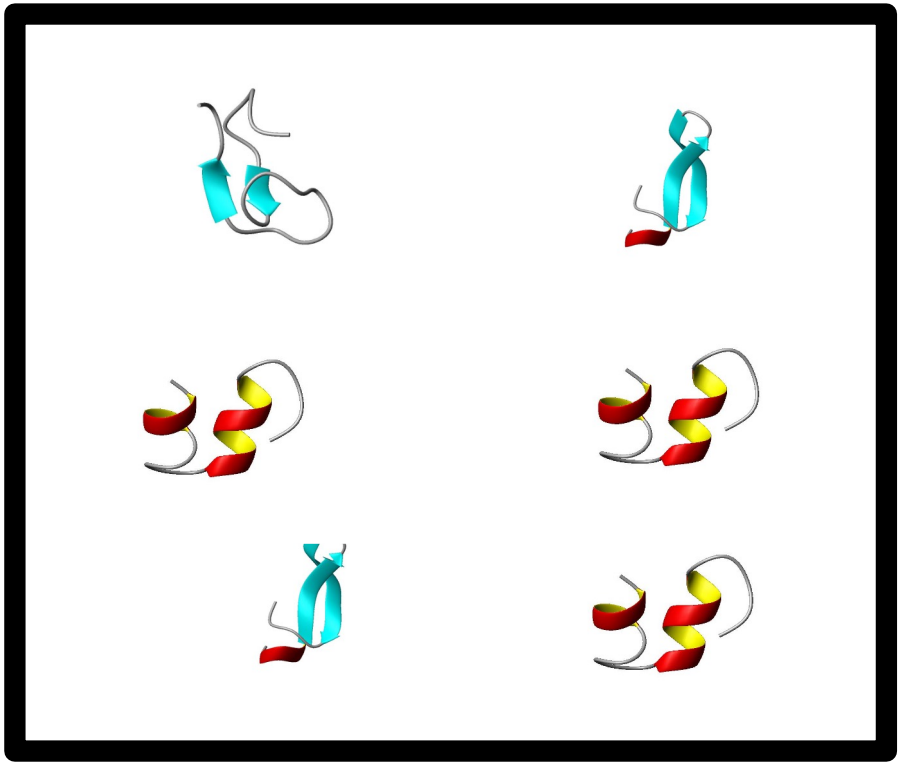


Molecular Simulation vs Experiment

1 molecule protein/
5000 water molecules



10^{20} protein molecules in
~ 10^{21} water molecules



Usual experimental condition
1 mmol protein (10^{-3} 6.023 10^{23} $molecules$ in 1 $liter\ water$)

Molecular Simulation vs Experiment

Molecular Simulations enable us to predict system properties through the generation of several conformations of the systems in such a way that accurate values of a property can be obtained.

Experimental measurements are made on macroscopic samples. An experiment measures an average property averaged over a large numbers of molecules (macroscopic sample), usually averaged over the time of measurement.

Molecular Simulation vs Experiment

The ergodic hypothesis states that the time average equals the ensemble average

$$\langle A \rangle_{ensemble} = \langle A \rangle_{time}$$

The basic idea is if one allows the system to evolve in time indefinitely, that system will eventually pass through all possible states

Molecular Simulation vs Experiment

The value of a **property** *A* , in a molecular dynamics simulation, is to determine a time average of A, which is expressed as

$$\langle A \rangle_{time} = \lim_{\tau \to \infty} \int_{t=0}^{\tau} A(p^N(t), r^N(t))\, dt \approx \frac{1}{M} \sum_{i=1}^{M} A_i(p^N, r^N)$$

we run a molecular simulation, in which conformations are generated according to the probability density

Molecular Simulation vs Experiment

One goal of a molecular simulation is to generate enough representative conformations such that this equality is satisfied. If this is the case, experimentally relevant information may then be calculated using a feasible amount of computer resources. Because the simulations are of fixed duration, one must be certain to sample a sufficient amount of phase space.

Different properties have different relaxation time

Note molecular dynamics simulations enable also time-dependent behavior of the systems

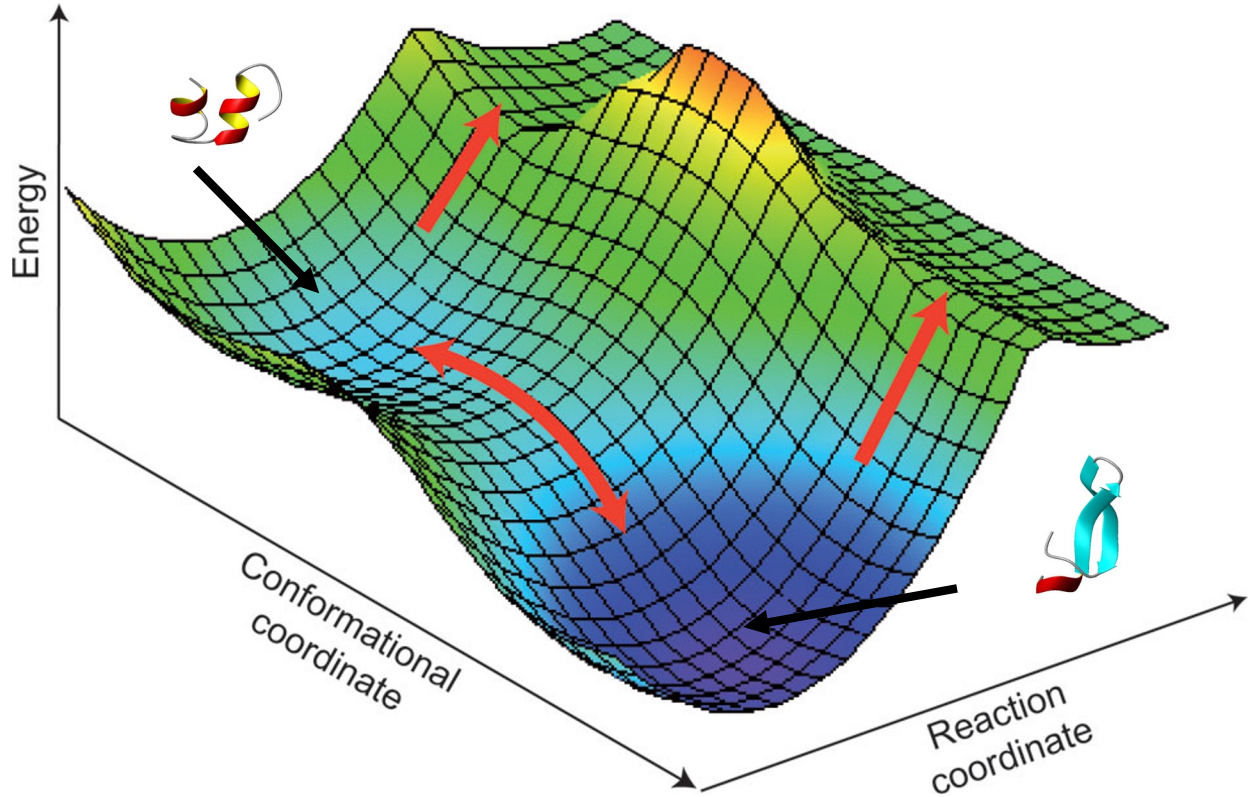
Challenge for biomolecular simulations

- Biophysical processes involve hundreds of thousands of atoms often in intricate interactions that are difficult to simply
- They span a wide range of time scales: primary events (e.i. photosynthesis) occur within ps, enzymatic and regulatory processes take milliseconds, and structural reorganizations may exceed seconds.
- The small driving forces that cause molecular changes results from large, opposing energetic effect. This required careful fine-tuning of the force fields that describe interatomic interactions.

One goal of a molecular simulation

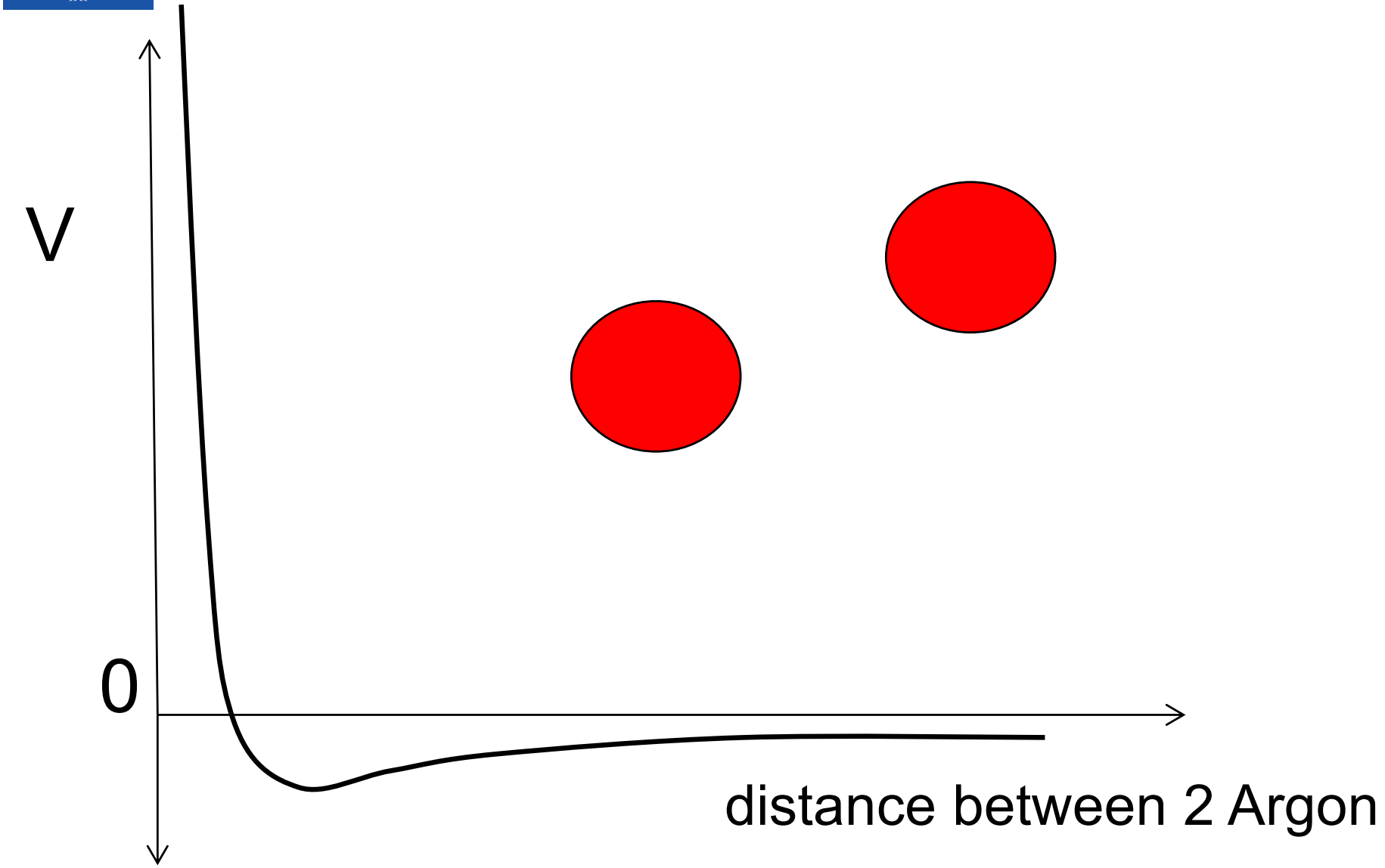
to generate enough representative conformations of the molecular system in such a way that accurate values of a property can be obtained.

Molecular simulation



A method to navigate the potential energy surface, to simulate the behavior of the molecular system

Potential energy surface of two Argon atoms



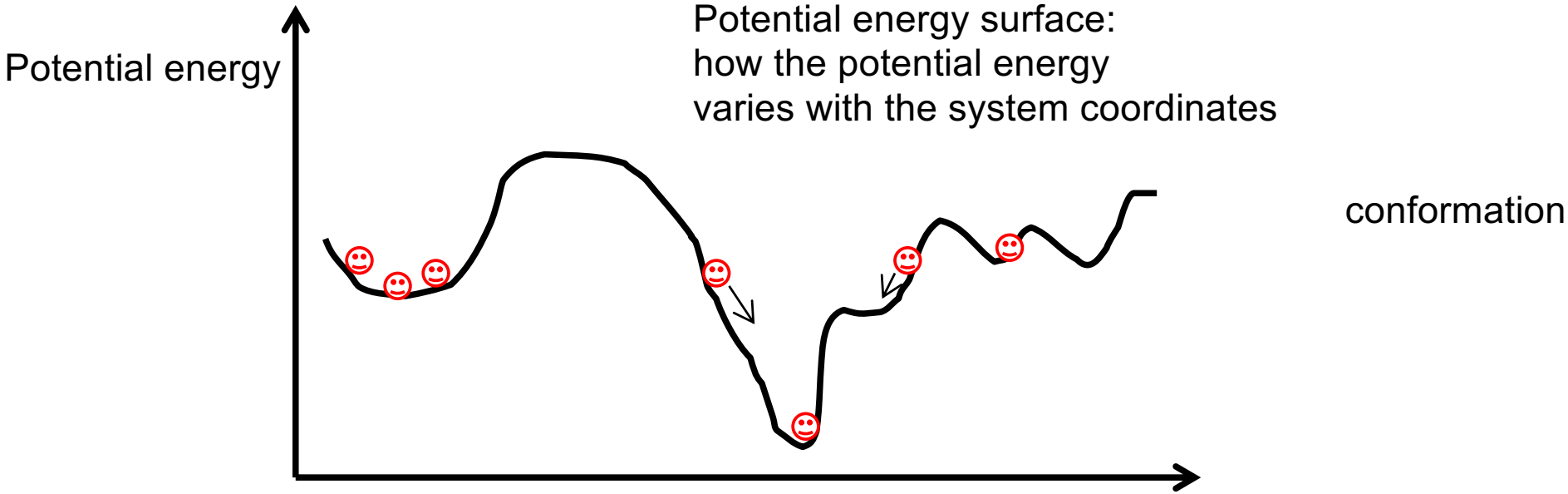
Potential energy surface

Assuming the energy of a molecule can be considered as function of nuclear positions only (Born-Oppenheimer approximation), potential energy surface is the way in which the energy of the molecular system varies with the coordinates

Thus, potential energy surface of a system depends on 3N-6 internal coordinates or 3N Cartesian coordinates.

=> Impossible to visualize for a biomolecule!!! except for very simple case when it is in function of only 1 or 2 coordinates.

Molecular dynamics simulation



Which methods for molecular simulations?

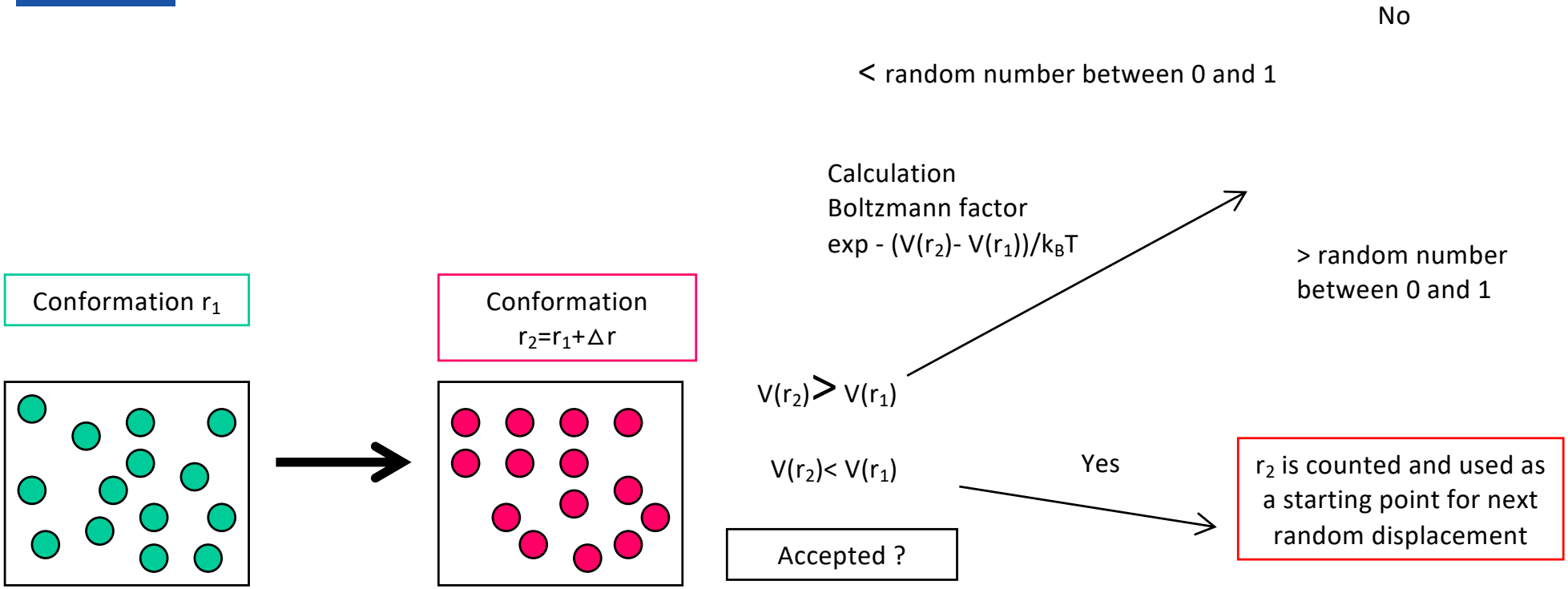
Monte-Carlo (MC) method

generates conformations by randomly generating new coordinates and energy criteria

Molecular Dynamics (MD) method

generates conformations by integrating Newton's law of motion

Monte Carlo simulation

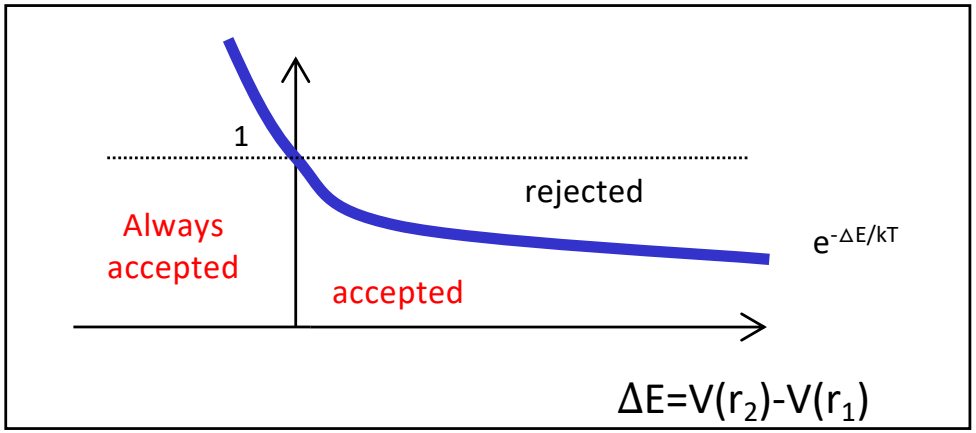


Generation of new configuration by randomly moving an atom or several atoms

Calculation of energy of new configuration

Monte Carlo simulation

Calculation
Boltzmann factor
exp $-(V(r_2)- V(r_1))/k_BT$



Low energy conformations have high chance to be accepted

$V(r_2) > V(r_1)$

$V(r_2) < V(r_1)$

Accepted ?

Calculation of energy of new configuration

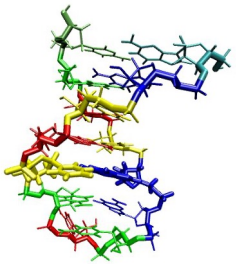
Always accepted

Monte Carlo simulation

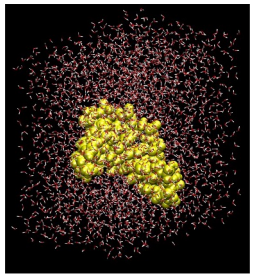
- This procedure generates a Boltzmann ensemble
- Most sampled configurations are relevant (low energy), while with other random sampling techniques many irrelevant configurations (high energy) are sampled.
- Efficiency depends on the step size Δr and the criteria used to accept or not the configurations
- This technique is rather inefficient for macro-molecular systems: a random displacement can generate high bond energy of the bonds of the displaced atom.

Monte Carlo simulation

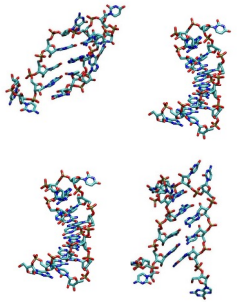
Starting structure



Condition



OUTPUT:
Ensemble of structures (called configurations)



thermodynamic properties

structural properties

Potential function to describe the molecular system

Monte Carlo method

Which methods for molecular simulations?

Monte-Carlo (MC) method

generates conformations by randomly generating new coordinates and energy criteria

Molecular Dynamics (MD) method

generates conformations by integrating Newton's law of motion

Newton's law of motion

Nature's law of motion

Newton's equation of motion

Force = Mass x Acceleration

or

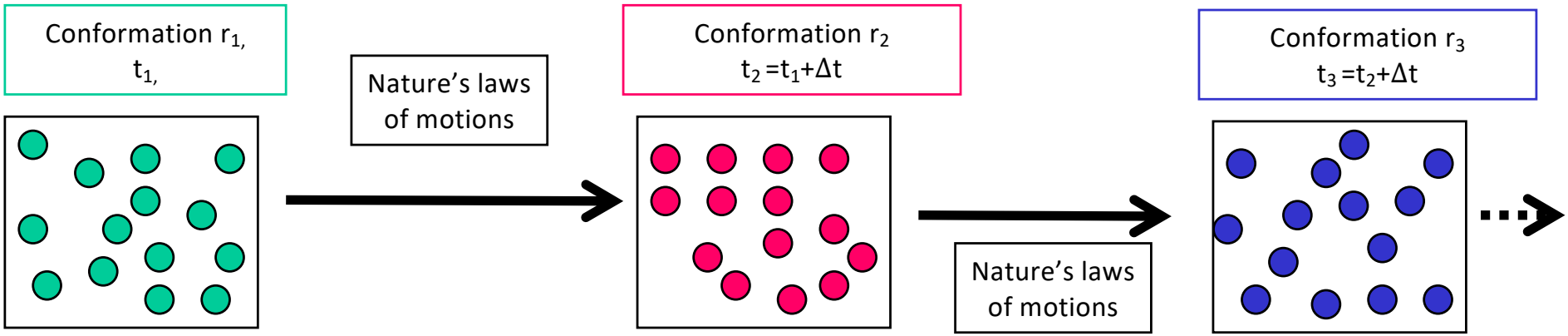
Acceleration = Force /Mass

$$\frac{d^2 r_i}{dt^2} = \frac{F_i}{m_i}$$

Where i refers to a particle i $(i = 1 \ldots\ldots \mathrm{N}_{\mathrm{particles}})$

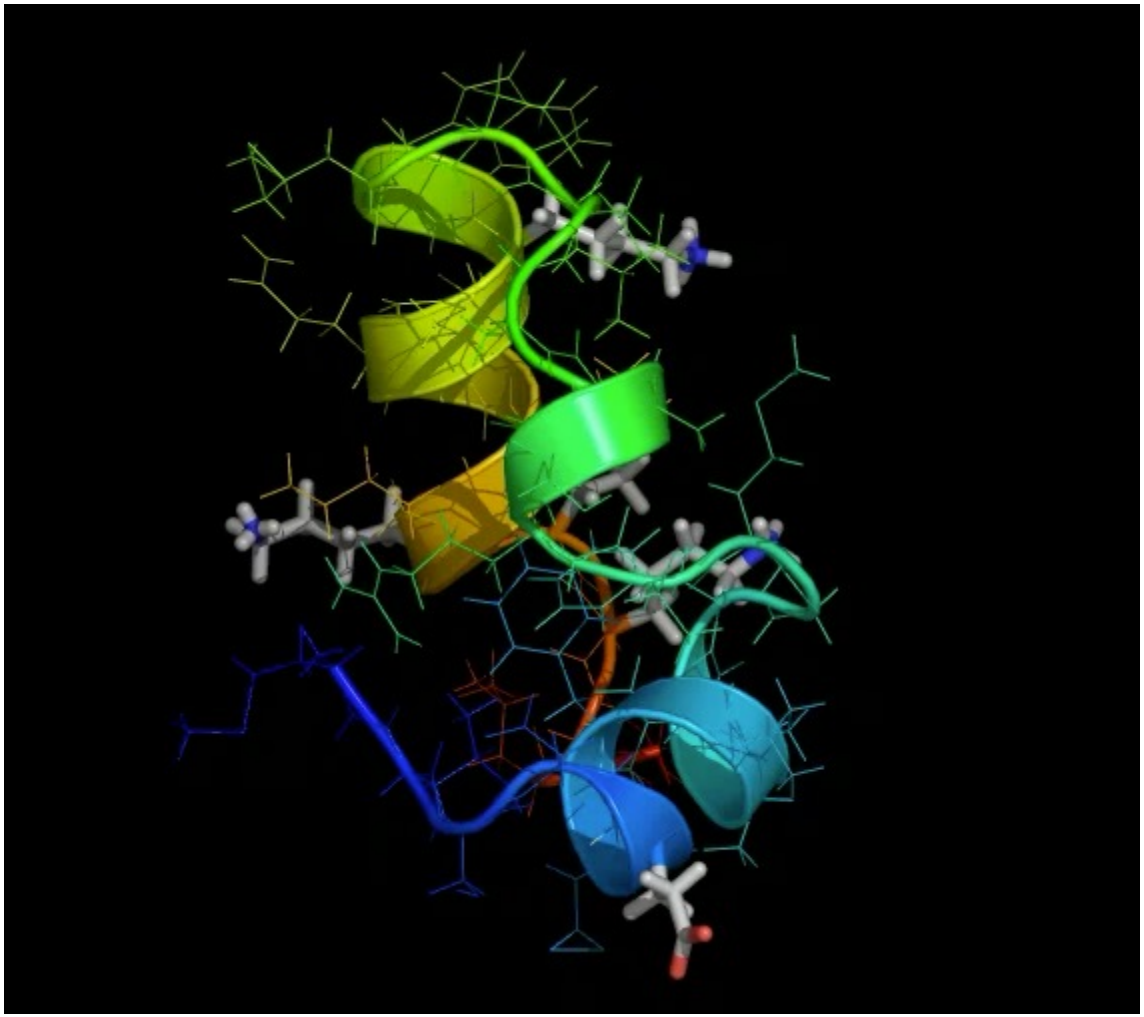
Molecular dynamics simulation

The Molecular Dynamics method generates configurations applying Nature's law of motion for atoms of a molecular system.



One small step for a human,many small steps for a computer

Phase space sampling



Newton's equation of motion

acceleration = force /mass

$$\frac{d^2 r_i}{dt^2} = \frac{F_i}{m_i}$$

force = -slope of the potential

$$F_i = -\frac{\partial}{\partial r_i} V(r_1, r_2 \ldots\ldots\ldots r_N)$$

$$i = 1 \ldots\ldots\ldots \mathrm{N}_{\mathrm{particles}}$$

One small step for a human,
many small steps for a computer

Molecular dynamics simulation

Nature's law of motion / Newton's equation of motion

Force = Mass * Acceleration

$$F_i\,(t) = m_i a_i(t) = m_i \frac{d^2 r_i}{dt^2}$$

Where i refers to a particle i $(i = 1 \ldots\ldots\ldots \mathrm{N}_{\mathrm{particles}})$,

$F_i\,(t)$ is the force acting on the particle i at time t

$a_i(t)$ is the acceleration on the particle i at time t

m_i is the mass of particle i

Difference between methods

- Monte Carlo (MC) simulations
- Molecular Dynamics (MD) simulations

FIVE points to think about

- generation an ensemble of conformations
- information on time dependence of properties
- contributions to the total energy
- role of the potential energy
- factors that regulate the efficiency of method

Molecular dynamics simulation

Nature's law of motion / Newton's equation of motion

Force = Mass * Acceleration

second-order differential equation

$$F_i\,(t) = m_i a_i(t) = m_i \frac{d^2 r_i}{dt^2}$$

$$F_i = -\nabla_i\, V(r_1, r_2 \ldots\ldots\ldots r_N)$$

$$i = 1 \ldots\ldots\ldots \mathrm{N}_{\mathrm{particles}}$$

Has to be solved numerically! Many dimension, and nonlinear function V (potential).

System of 2:nd order partial differential equations

Molecular dynamics simulation

Nature's law of motion / Newton's equation of motion

Force = Mass * Acceleration

$$F_i\,(t) = m_i a_i(t) = m_i \frac{d^2 r_i}{dt^2}$$

$$F_i = -\nabla_i\, V(r_1, r_2 \ldots\ldots\ldots r_N)$$

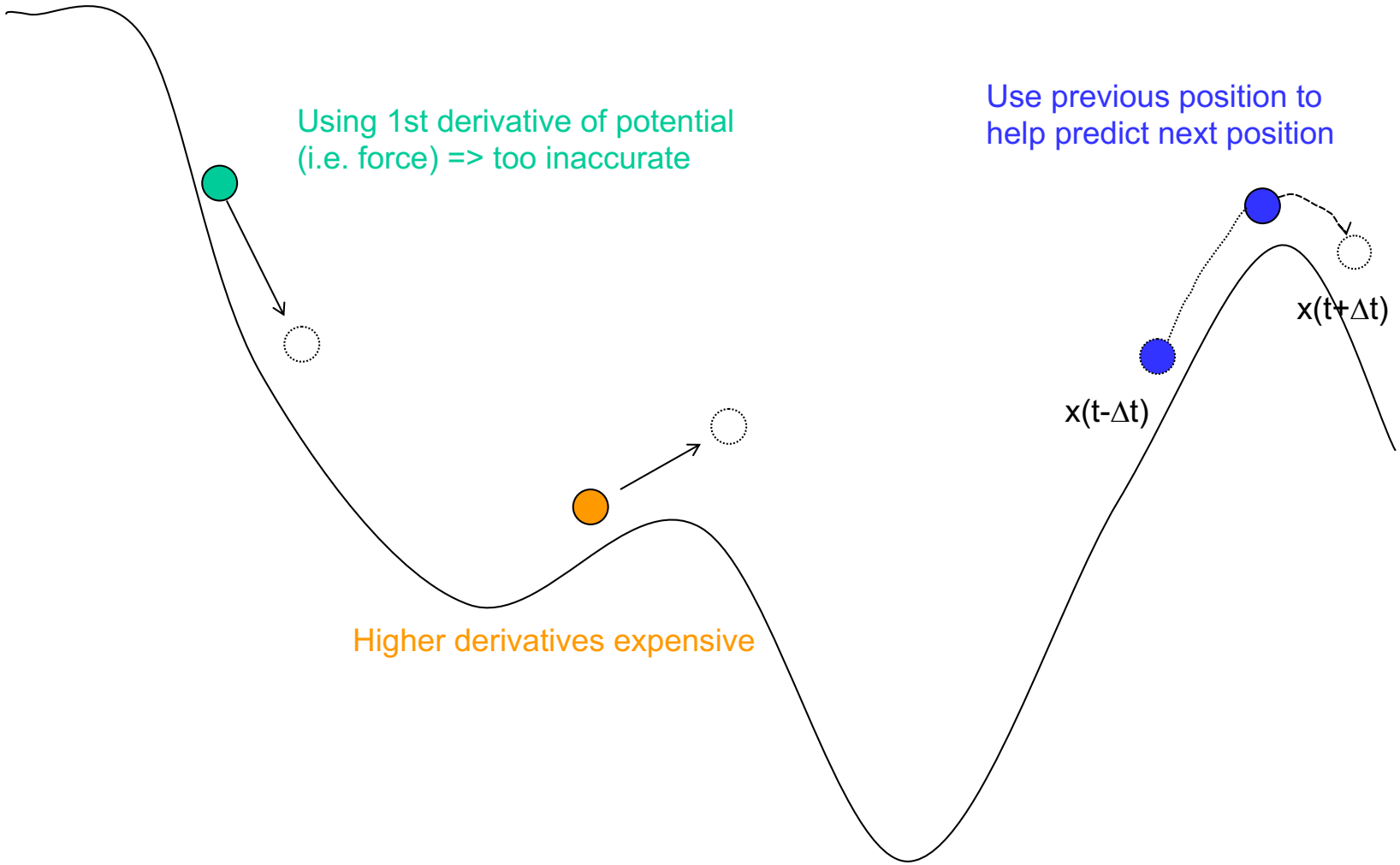
$$i = 1 \ldots\ldots\ldots \mathrm{N}_{\mathrm{particles}}$$

Has to be solved numerically! Many dimension, and nonlinear function V (potential). System of 2:nd order partial differential equations

Solution:

simple algorithm to integrate it in small time steps

Algorithm for molecular dynamics



Algorithm for molecular dynamics

All the algorithms used assume that positions and dynamical properties can be approximated as Taylor series expansions:

$$r(t+\delta t) = r(t) + v(t)\delta t + a(t)\,\delta t^2/2 + b(t)\,\delta t^3/6 + c(t)\,\delta t^4/24\ldots$$

$$v(t+\delta t) = v(t) + a(t)\delta t + b(t)\,\delta t^2/2 + c(t)\,\delta t^3/6\ldots$$

$$\mathrm{a}(t+\delta t) = a(t) + b(t)\delta t + \mathrm{c}(t)\,\delta t^2/2 + ..$$

$$\mathrm{b}(t+\delta t) = b(t) + c(t)\delta t + ..$$

Algorithm for molecular dynamics

Verlet's algorithm:

$$r(t+\delta t) = 2\mathrm{r}(t) - r(t-\delta t) + a(t)\delta t^2$$

The resulting **trajectory** r(t) contains (almost) all information about the system. Velocities are most of the time less interesting.

Algorithm for molecular dynamics

Leap-frog's algorithm

(one of the variations on the Verlet algorithn)

$$r(t+\delta t) = r(t) + v(t+\frac{\delta t}{2})\delta t$$

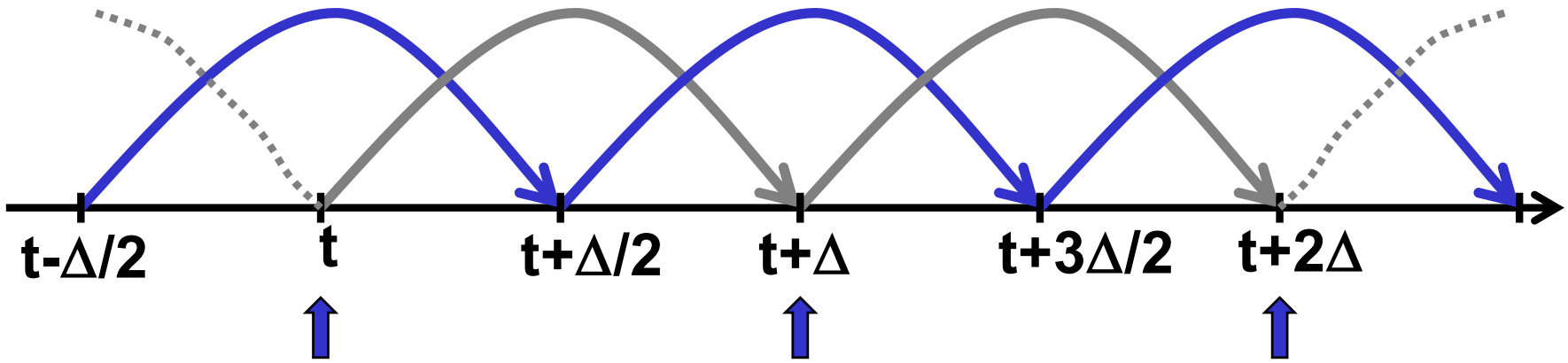
$$v(t+\frac{\delta t}{2}) = v(t-\frac{\delta t}{2}) + a(t)\delta t$$

mathematically same but numerically better.

the leap-frog integrator

Velocities propagation

Coordinates propagation



Force calculation

- Explicit inclusion of velocities, but: coordinates and velocities are not available simultaneously (velocities at time t must be back calculated)
- Does not require difference of large numbers

Exercise

Write the two possible expressions for the velocity v(t)
in Verlet algorithm, knowning that:

- the velvet algorithm uses the positions to calculate the velocity

- the velocity can be estimated at time t or time t+1/2dt



Algorithm for molecular dynamics

Exercise

Think at possible drawback of Verlet and leap-frog algorithm

Material on the web

Algorithm for molecular dynamics

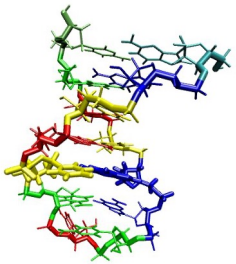
Points to discuss with your neighborghs

- Leap-frog and Verlet algorithms are self-starting algorithm or not

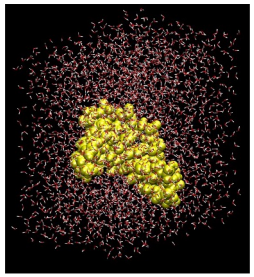
- Information that the leap-frog and Verlet algorithms need to store to calculate the position at the time t+dt

Molecular dynamics simulation

Starting structure



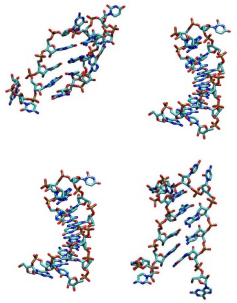
Condition



Potential function to describe the molecular system

Molecular dynamics method

OUTPUT: Ensemble of structures (called configurations)



thermodynamic properties

structural properties

kinetic properties

dynamic information

Molecular Dynamics Simulation

The total energy of a system of N atoms of mass m_i , (i=1,.....N), is given by

$$H(\mathbf{p},\mathbf{r}) = K(\mathbf{p}) + V(\mathbf{r})$$

$K(\mathbf{p})$: kinetic energy

$V(\mathbf{r})$: potential energy

$$K(\mathbf{p}) = \Sigma i\ \mathbf{p}_i^2 / 2\ m_i \qquad V(\mathbf{r}) = V(r_1, r_2 \ldots)$$

p_i (= $v_i * m_i$) is momenta, m_i is the mass, r_i is the position of atoms i

Classical approximation

Particles ≡ Atoms (no electrons are included)



THANK YOU